

SUMMARY TABLE: ESSENTIAL MATHEMATICS GRADE 10

Sub-Strand 3.1: Statistics 1 — Teacher Reference (Pre-filled)

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Sub-Strand	Sub-Strand 3.1: Statistics 1
Driving Question	DRIVING QUESTION: How can we turn raw data from our school and community into statistics and graphs that help Mama Wanjiku — and other Kenyan decision-makers — solve real problems?

INSTRUCTIONS

FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 What Is the Data Telling Us? — Launching the Canteen Queue Puzzle	Learners examined a raw list of 30 meal orders from Mama Wanjiku's canteen and experienced firsthand how difficult it is to answer simple questions (which meal is most popular? how many ugali orders?) from unorganised data — even when they were allowed to count freely. Different groups reached different totals for the same list.	Raw data is essentially invisible information. A frequency distribution table — built using tally marks — organises data so that patterns (such as ugali being the most popular meal) become immediately visible and verifiable. Key vocabulary introduced: raw data, tally, frequency, frequency distribution table.	Mama Wanjiku's memory-based guessing fails because the human mind cannot reliably track and compare frequencies across many transactions without a systematic recording tool. The frequency distribution table is the first step in transforming raw data into a decision-making tool — exactly what Mama Wanjiku needs to know how much of each meal to cook each day.
Lesson 2 From Chaos to Clarity: Collecting Raw Data to Help Mama Wanjiku	Learners examined a 'raw data wall' of 120 unsorted meal choices from Mama Wanjiku's canteen and attempted to count ugali entries from a 30-item data strip — experiencing first-hand how unorganised data leads to inconsistent and unreliable counts.	Ungrouped (raw) data is a list of individual data entries that has not yet been sorted or counted. A tally sheet organises data collection by using tally marks against pre-defined categories, making counting faster, more accurate, and reproducible. Designing a good tally sheet requires mutually exclusive categories and a clear research question.	Groups designed and used their own tally sheets to collect at least 30 real data entries from classmates on a school-relevant question (favourite meal, transport mode, or sport). They connected this process directly to Mama Wanjiku's problem: she never had a tally sheet — she relied on memory — which is why her guesses about meal popularity were so wrong.
Lesson 3 Finding the Middle Ground — Mean, Mode and Median for Mama Wanjiku	Learners observed that the five daily ugali sales totals (54, 60, 57, 60, 59) are not all the same — they vary from day to day — and that the number 60 appears twice, making it stand out as the most common	Three measures of central tendency summarise ungrouped data in different but complementary ways: the MEAN ($290 \div 5 = 58$) gives the mathematical average; the MODE (60) identifies the most frequently	Mama Wanjiku's memory-based guesses failed because human memory tends to anchor on recent or dramatic events rather than calculating a true average across all days. The mean of 58 portions per day is a

	(peak) day. They also observed that ordering the data (54, 57, 59, 60, 60) makes both the middle value and the repeated value immediately visible.	occurring value; and the MEDIAN — found at position $(n+1)/2 = 3\text{rd value} = 59$ — gives the middle value of the ordered dataset. Each measure answers Mama Wanjiku's planning question slightly differently: mean gives a balanced daily target, mode reveals the peak demand day, and median offers a robust middle estimate.	precise, reproducible number — derived from the full week's evidence — that removes the distortion of memory. The mode of 60 further reveals that on the most common type of day she needs even more portions than the mean suggests. Together, these statistics transform the noisy, unorganised raw data from Lesson 2 into a concrete, mathematically justified daily cooking target — exactly the kind of decision-support tool that Mama Wanjiku, and any community decision-maker, needs.
Lesson 4 From Chaos to Clarity: Building Frequency Distribution Tables	We observed that when we sorted our raw list of 120 canteen meal orders into a frequency distribution table — listing each meal and tallying how many times it appeared — the data stopped looking like noise and started showing a clear pattern. In every group's table, ugali with sukuma wiki had the highest frequency (around 55–60 out of 120), while githeri had the lowest. The same pattern appeared in the transport data the teacher modelled: once tallied, the most and least common modes of transport became immediately obvious.	We learned that a frequency distribution table organises ungrouped raw data by recording each possible value and its frequency (how often it occurs), using tally marks grouped in fives for easy counting. The sum of all frequencies must equal the total number of data entries — this is our accuracy check. Organisation reveals patterns that are invisible in a raw list, and it was this kind of organisation that Mama Wanjiku was missing when she relied on memory alone.	We can now explain why Mama Wanjiku's memory-based guesses failed: she was trying to process 120 individual orders mentally, without any system for counting or recording. A frequency distribution table does what memory cannot — it counts every single entry without bias, groups them by category, and displays the pattern clearly. With a completed table, Mama Wanjiku can see at a glance that she needs roughly 58 portions of ugali with sukuma wiki every day and far fewer portions of githeri, allowing her to order ingredients accurately, reduce waste, and ensure every learner gets served. The table does not yet tell us what a 'typical' day looks like as a single number — that is the question we carry into Lesson 5.
Lesson 5 Picture This — Representing Mama Wanjiku's Data as Bar Graphs, Line Graphs and Pie Charts	Learners constructed three graphs from Mama Wanjiku's canteen frequency data — a bar graph showing meal popularity, a pie chart showing the proportion each meal represents out of 150 total orders, and a line graph showing daily ugali sales across Monday to Friday. In the gallery walk they observed that different graph types made different patterns visible: the bar graph made size comparisons easy, the pie chart communicated proportions (ugali = 37%), and the line graph was the only one to reveal the time-based trend in ugali demand.	A bar graph uses bar heights to compare frequencies between categories and is ideal for showing which canteen meal is most or least popular. A pie chart uses sector angles — calculated as $(\text{frequency} \div \text{total}) \times 360^\circ$ — to show each category as a proportion of the whole, making it clear that ugali accounts for 37% of all orders. A line graph connects plotted data points across a time sequence and is the only graph type that can show whether ugali demand is rising, falling or stable across the school week. The choice of graph type depends on the question being asked, not personal preference.	Mama Wanjiku's memory-based guesses failed because the human mind cannot accurately track frequency, proportion and time-trend simultaneously across hundreds of transactions. Each graph type makes one of those three dimensions instantly visible: the bar graph would have shown her at a glance that githeri is ordered four times less often than ugali; the pie chart would have shown that nearly two in every five orders are for ugali; and the line graph would have confirmed that ugali demand is consistently high every day of the week, not randomly variable. Together, these three visual tools transform the same frequency table into three different decision-making instruments

			— exactly the kind of organised, trustworthy information that Mama Wanjiku was missing when she ran out of ugali and was left with unsold githeri.
Lesson 6 Measures of Central Tendency: Mean and Mode — Finding the 'Typical' Meal for Mama Wanjiku	Mama Wanjiku's five-day ugali sales data (54, 60, 57, 60, 59 portions) and Kericho monthly rainfall figures (Jan–Jun) — both presented as raw lists of ungrouped values.	The mean ($\text{sum} \div \text{number of values} = 290 \div 5 = 58$) and the mode (most frequent value = 60) are two different numerical summaries of the same data set; each answers a slightly different question about what is 'typical,' and mode only works meaningfully when values repeat.	Mean and mode are both measures of central tendency for ungrouped data. They can give different answers because mean accounts for the size of every value while mode counts only frequency of occurrence. For Mama Wanjiku, mode (60) tells her the most common single-day demand, making it useful for daily stock ordering, while mean (58) smooths out day-to-day variation, making it useful for weekly budgeting. The Kericho rainfall data (mean = 147.5 mm/month, no mode) shows that mode is most powerful for discrete data where repetition occurs — and that a third measure of central tendency (median) may sometimes be needed.
Lesson 7 The Middle Value: Median — Finding the Centre of Mama Wanjiku's Data	Learners physically ordered 7 daily ugali-portion values from Mama Wanjiku's canteen data using a Human Number Line activity, and identified the middle value (58). They then observed what happens when an 8th value (56) is added, creating an even-numbered dataset with two middle values (57 and 58), requiring them to calculate the average (57.5). They observed that the median barely changed when the sports-day outlier of 92 portions was added to the dataset.	Learners learned that the MEDIAN is the middle value of a dataset that has been arranged in ascending (or descending) order. For an odd number of values (n), the median is at position $(n+1)/2$. For an even number of values, the median is the average of the two middle values. The median is a ROBUST measure — unlike the mean, it is not significantly affected by extreme values (outliers) such as an unusually busy sports-day at the canteen. For Mama Wanjiku's 7-day ugali data, the median is 57 portions per day.	Learners explained that Mama Wanjiku's memory-based guesses failed partly because her memory was anchored to the most vivid or extreme days (like sports day with 92 portions), which is analogous to the mean being pulled upward by outliers. The median gives a more reliable daily planning target (57 portions) because it reflects the true MIDDLE of her typical trading days, unaffected by exceptional events. Learners updated their cumulative Statistical Models with the median arrow and wrote personalised two-sentence recommendations to Mama Wanjiku, integrating mean, mode, and median into a coherent statistical argument.
Lesson 8 The Full Picture — Final Explanation: Giving Mama Wanjiku a Complete Statistical Report	Learners observed the complete contrast between the original raw, unorganised list of 120 canteen meal orders from Lesson 1 and their fully built statistical model from Lessons 1–7, seeing in a single side-by-side comparison exactly how each tool (tally sheet → frequency table → mean/mode/median → graphs) added a layer of organisation, summarisation or	Learners consolidated that frequency distribution tables, mean, mode, median, bar graphs, line graphs and pie charts are not isolated skills but a coherent, sequential statistical process: raw data must first be organised (frequency table), then numerically summarised (measures of central tendency), then visually represented (graphs), before any evidence-based	The anchoring phenomenon is now fully explained: Mama Wanjiku's memory-based guesses failed because raw, unorganised data is essentially invisible information — the human memory cannot reliably detect statistical patterns across 120 data points without systematic tools. A frequency distribution table made the pattern visible; measures of central tendency gave her a

	visualisation to transform invisible data into an actionable recommendation for Mama Wanjiku.	decision can be made. They also learned that the same process scales beyond Mama Wanjiku's canteen to any real Kenyan decision-maker — a maize farmer in the Rift Valley, a matatu operator in Nairobi, a nurse tracking disease cases near Lake Victoria — who needs to transform raw data into knowledge.	specific, defensible planning target (prepare approximately 57–60 portions of ugali per day); and graphs allowed her to communicate the finding to others at a glance. The Canteen Queue Puzzle is solved — and the statistical toolkit that solved it is the same one that can be applied to any data-rich problem in Kenyan community life.
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ATTRIBUTION AND LICENSING

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